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Betsi Cadwaladr
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ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

ESTABLISHING EPIDURAL ANAESTHESIA FOR LABOUR ANALGESIA

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1. Introduction and Purpose

Epidural anaesthesia has been available in the UK for labour analgesia since the late 1960's. It is a more effective form of analgesia compared with opiates and Entonox. However it is associated with risks such as post dural puncture headache (1:100) and permanent nerve damage (1:24,000 at 6 months).

2. Scope

This guideline is intended as a reference for anaesthetic trainees seeking a recommended technique for epidural anaesthesia for labour analgesia.

3. Points to consider

3.1 Patient Preparation

- Ensure the midwife has provided the patient with the Epidural Information Card [2]
- Full preoperative assessment of the patient
- Indications for epidural analgesia include :
    i. Maternal request
    ii. Multiple pregnancy or malpresentation (ie. increased risk of operative delivery)
    iii. Pre-eclampsia
    iv. Maternal cardiovascular disease (eg. IHD or regurgitant valvular lesions)
    v. Morbid obesity (to try and avoid a GA if operative delivery necessary)
- Ensure no contraindication to epidural :
    i. Patient refusal
    ii. Lack of midwifery supervision
    iii. Thrombocytopaenia (platelet count < 80)
    iv. Coagulopathy (INR >1.4) or patient on anticoagulation therapy
    v. Uncorrected hypovolaemia
    vi. Local sepsis
    vii. Systemic sepsis

Criteria for the diagnosis of systemic sepsis :

- *Clinical suspicion of sepsis* :

History - e.g diarrhoea, vomiting, UTI, chest infection chorioamnionitis etc.

& Clinical signs - Heart rate >100 / minute
RR >20 / minute
+ Temp. <36 ºC or > 38 ºC *

or - *Temp. >38 ºC* in the absence of an obvious non-infective cause

or - *WCC > 30,000 / mm³* in the absence of an obvious non-infective cause

or - *CRP > 40* in the absence of an obvious non-infective cause[1]

*** All temperature measurements should be undertaken for accuracy using a temporal artery scanning temperature probe**

(NB. Chronic backpain does not contraindicate epidural anaesthesia, but scarring of the epidural space e.g following spinal surgery may limit the effectiveness of epidural analgesia and increase the risk of inadvertent dural puncture)

- Discussion with the patient of:

    a) Benefits of an epidural:

        i. Better quality labour analgesia than opiates or Entonox

        ii. Can be topped up to give anaesthesia for Caesarean Section, instrumental delivery, manual removal of placenta or repair of perineal tears

        iii. In pre-eclampsia it improves uteroplacental bloodflow and reduces plasma catecholamines and stress response with consequent better control of blood pressure

    b) Risks of an epidural:

        i. Post dural puncture headache (1:100)

        ii. Nausea and vomiting

iii. Failure to provide effective analgesia for labour (1:8)

Failure to produce sufficient analgesia on topping up for Caesarean Section thus requiring a conversion to general anaesthesia (1:20)

iv. Nerve damage (Temporary = 1:1,000,
Permanent = 1:24,000 incidence at 6 months)

c) Epidural procedure

d) Explanation of what sensations she will feel once epidural is functioning

- Obtain verbal consent for the epidural and document this on the epidural chart
- Discuss local anaesthetic delivery ie. Programmed Intermittent Bolus (PIB) plus Patient Controlled Epidural Analgesia (PCEA)

3.2 Equipment and Staff

- Ensure presence of trained midwife to monitor the epidural block once established
- Check all equipment is present to perform epidural block
- Ensure metaraminol and ephedrine are at hand

3.3 Technique

- Attach monitoring :
  - NIBP monitor to record a baseline blood pressure and pulse
  - Cardiotocography (CTG) monitor if not already in place to record a baseline fetal heart rate
- Site a **16G** (or 14G) iv cannula
- A routine fluid preload is not required, however if the parturient is dehydrated a preload volume of **500ml Hartmanns solution** should be administered. Do not preload in severe pre-eclampsia
- Place patient in **Sitting** (can be cross-legged) or lateral position
- Use full aseptic technique :

a) Gown, mask and gloves

b) Spray the patient's back twice from the mid thoracic area to the upper natal cleft, and over to the flanks using **Derma Spray** (0.5% chlorhexidine in 70%

alcohol). Allow the alcohol to dry each time.

- Skin and ligament infiltration with 1% Lignocaine at **L3/4** or L4/5 interspace

- 16G Tuohy needle (8cm) used (11cm Tuohy needle available for super morbidly obese) and inserted just into the ligament, followed by removal of the stylet

- Epidural space located using loss of resistance to saline technique. This is nowadays preferred to LOR to air. Remember to always use the LOR syringe and not an ordinary 10 ml syringe (high resistance)

- Epidural catheter is threaded and left 5 cm in the epidural space (7cm in morbid obesity)

- Check that no blood or CSF can be aspirated from the epidural catheter

- **The falling meniscus sign is a recognised and useful indicator of correct epidural catheter placement, <u>but should never be used as a substitute for administering an epidural test dose</u>**

- Epidural catheter secured with adhesive clip, Tegaderm and Mefix dressings

- Draw up 15mls 0.1% levobupivacaine + 2 micrograms fentanyl from the standard epidural infusion mixture bag into a 20ml NRFit syringe **<u>using an aseptic technique</u>**

- **EPIDURAL TEST DOSE PROTOCOL** : **Administer a test dose of <u>10mls 0.1% levobupivacaine + 2 micrograms fentanyl</u> slowly over 2 minutes**

- **Immediately following administration of the epidural test dose, commence maternal pulse and blood pressure measurements at 5 minute intervals for 20 minutes. Continuous CTG monitoring should also be in place. If the anaesthetist were to confirm that the epidural test dose is positive for intrathecal placement, then the frequency of maternal pulse and blood pressure measurements should be increased to 2 minute intervals for a period of 10 minutes**

- **The anaesthetist and anaesthetic assistant must remain in the labour room for the first 5 minute period following the administration of the epidural test dose, and should closely observe the patient for any developing signs of a subarachnoid block**

- **<u>Wait a full 5 minute period following the test dose</u>, and if at this point there is no evidence of a subarachnoid block (high sensory & dense motor block) or intravascular injection (perioral tingling, lightheadedness, tinnitus etc.), then proceed to administer a loading dose of <u>5ml 0.1% levobupivacaine + 2 micrograms fentanyl</u>**

- The midwife will programme the Sapphire epidural infusion pump by selecting the PIB / PCEA programme. Check the programme, which should read as follows :

  Drug : 0.1% Levobupivacaine + 2mcg/ml fentanyl

  Dose : PIB = 10ml
  PCEA = 7ml

  Lockout : PIB = 60 minutes
  PCEA = 20 minutes

- Connect the epidural infusion set to the epidural filter and start the epidural infusion pump

- **Please be aware that the epidural infusion pump will administer it's first 10ml PIB dose soon after it is started, hence the recommendation for a reduced 5ml loading dose to be administered if the epidural infusion pump is to be commenced as soon as the loading dose has been administered.**

- Document the epidural procedure on the front page of the epidural chart and select the PIB / PCEA program (which should always be the default choice)

- Prescribe the Epidural Infusion for Labour order set on ePMA

- At 20 – 30 minutes post epidural establishing dose, assess the level of sensory block (temperature) using Ethyl Chloride spray and motor block using the Bromage Scale. Aim for T10 block bilaterally with minimal motor blockade

- Return to the patient at regular intervals during labour to assess the quality of epidural analgesia and liaise with midwife regarding any problems with the epidural.

3.4 Monitoring

- Irrespective of the epidural pump program chosen (PIB / PCEA or continuous infusion) the monitoring regime outlined below should be adhered to[6].

- Perform routine haemodynamic measurements (pulse & NIBP) every 5 minutes for 20 minutes following the epidural administration of :

  a) The initial establishing local anaesthetic / opiate bolus

  b) Any bolus dose **greater than** 10mg Levobupivacaine
  (ie greater than 10 mls 0.1% Levobupivacaine or equivalent)

- At **30 minute intervals** during the maintenance of epidural analgesia the midwife should monitor pulse & blood pressure. This should be performed more frequently if the patient is not cardiovascularly stable

- At **hourly intervals** during the maintenance of epidural analgesia the midwife should monitor :

    a) Respiratory Rate

    b) Height of sensory block (cold sensation) – including noting any missed segments

    c) Motor block

    d) Sedation score

    e) Pain score

    f) Epidural site

    g) Degree of bladder distension (if urinary catheter not in situ)

    h) Pressure areas

3.5 Discontinuation of the epidural

- Following satisfactory conclusion of third stage of delivery and repair of any perineal tears, the midwife should :

    a) Stop the epidural infusion

    b) Disconnect the epidural infusion and cap the epidural filter

    c) Aseptically (using sterile gloves) remove the epidural catheter and dressings and make sure the tip is intact. If the epidural catheter is stuck try flexing the patient more. If it still cannot be removed then the anaesthetist should be called. NEVER USE EXCESSIVE TRACTION TO REMOVE AN EPIDURAL CATHETER AS IT MAY BREAK

    d) Midwife should document that the tip of the epidural catheter was intact

    e) Place a plaster over the epidural site

    f) Monitor offset of epidural block for 6 hours after removing the epidural catheter

    g) Withhold enoxaparin dose for 4 hours following removal of the epidural catheter

DO NOT REMOVE THE EPIDURAL CATHETER IF THE PLATELET LEVEL < 80 OR THE PATIENT HAS A COAGULOPATHY (eg severe pre-eclampsia ,HELLP, APH or PPH)

3.6 Methods of epidural infusion delivery

- Programmed Intermittent Bolus (PIB)

  The epidural pump delivers a bolus of a set volume at a set interval e.g instead of a 10ml/hr infusion, the pump delivers a regular 10 ml bolus at hourly intervals. PIB results in a greater and more uniform spread of local anaesthetic compared to continuous epidural infusion (CEI) due to higher injectate pressures. The quality of analgesia provided by PIB is similar to CEI, however this method of delivery of epidural analgesia has been shown to reduces local anaesthetic consumption & motor block, reduces the number of clinician top up boluses required and may improve patient satisfaction[5]. PIB can be used with or without a Patient Controlled Epidural Analgesia (PCEA) function.

- Patient Controlled Epidural Analgesia (PCEA)

  The patient initiates delivery of a bolus of local anaesthetic epidurally. The volume of the bolus and lockout interval is pre-programmed. As with PIB the quality of analgesia provided by PCEA is similar to CEI. This method of delivery of epidural analgesia has also been shown to reduces local anaesthetic consumption & motor block, reduces the number of clinician top up boluses required and may improve patient satisfaction[3,4]. Fewer but larger volume PCEA doses (> 5ml) have been found to provide better analgesia than more frequent smaller doses (due to the local anaesthetic spread effect). PCEA can be used with or without PIB or a continuous background infusion.

- Continuous Epidural Infusion (CEI)

  A continuous infusion of local anaesthetic / opiate mix is administered epidurally. CEI is associated with an increase in local anaesthetic consumption and motor block compared with an intermittent bolus techniques, due to a poorer spread of local anaesthetic[3,5].

  Therefore in summary, epidural infusion delivery methods in order of best maternal satisfaction scores (& lowest LA consumption) are as follows :

  PCEA + PIB > PCEA + background CEI > PCEA alone > CEI

  For this reason PIB + PCEA is therefore the preferred method of delivery of epidural LA in our maternity unit.

3.7 Sapphire Epidural Infusion Pump Programmes

The Sapphire pump has two obstetric Epidural infusion programmes :

a) Combined PIB & PCEA programme (Obs PIB PCEA)

b) Continuous infusion programme (Obs Continuous)

- **Setting up the Sapphire epidural pump for PIB / PCEA Programme**

  Ensure the external PCEA handset is plugged in (port on the bottom, ensure red spot is inserted at 12'oclock)

  Switch the pump '**ON**' (yellow button on the bottom right)

  Screen displays '**Obs Continuous' or 'Obs PIB PCEA'** (depending on which protocol was used on last patient).

  Choose **'ACCEPT'** or '**CHANGE**' to '**Obs PIB PCEA**' and '**OK**' to continue

  Press '**Epidural'** on the screen

  Press **'Intermittent'** on the screen

  Press **'PreSet Programmes'** (bottom) of screen

  'New Patient' - Press 'Yes'

  Press **'Obs PIB PCEA'** on the screen

  Protocol is displayed on the screen

  Review pump settings : Program Name: Obs PIB PCEA

  Drug Name : 0.1 Levobupivacaine + 2mcg/ml fentanyl

  VTBI (reservoir volume) : 100ml (default) (Range 0-1000ml)

  PIB (Intermittent) dose : 10 ml (not adjustable)

  PIB dose Interval : 1 hour (not adjustable)

  Continuous Rate : 0ml/hr

  PCEA (demand) bolus : 5 ml (default) (Range : 5 – 7 ml)

  PCEA (demand) bolus lockout : 20 minutes (not adjustable)

  Press '**Next**'

  Press **'Ok'**

  'Use bolus handle' – Press **'Yes'**

Press button on the handle to proceed

Do you want to PRIME before you start?

Choose **'PRIME'** or '**START'**

Delivery history' provides data on PIB and PCEA usage

To change settings

Press **'unlock to patient'** (bottom of screen)

Password **5551**

'**Request pause'** and '**OK**'

Press **'View/Edit'** (bottom of screen)

VTBI (10-100mls)

'Clinican bolus' is not available when using PIB/ PCEA programme

Input the required volume

Press '**OK**'

Press '**Request Continue'** (bottom of screen)

- **Setting up the Sapphire epidural pump for a Continuous infusion Programme**

  Ensure the external PCEA handset in plugged in (port on the bottom, ensure red spot is inserted at 12'oclock)

  Switch the pump '**ON**' (yellow button on the bottom right)

  Screen displays '**Obs Continuous' or 'Obs PIB PCEA'** (depending on which protocol was used on last patient).

  ACCEPT or **CHANGE** to '**Obs Continuous'**

  Press '**Epidural**' on the screen

  Press '**PCEA**' on the screen

'PreSet Programmes' (bottom) of screen

'New Patient' - Press **'Yes'**

Press **'Obs Continuous'** on the screen

Review pump settings : Program Name: Obs Continuous

Drug Name : 0.1 Levobupivacaine + 2mcg/ml fentanyl

VTBI (reservoir volume) : 100ml (default) (Range 0-1000ml)

Continuous Rate : 10ml/hr (default)
(Adjustable range : 10 – 15 ml / hr)

"Clinician bolus" function is only available when using the continuous infusion programme. The settings for clinician (midwife) bolus are as follows :

Clinician (midwife) bolus volume : 10 ml (Adjustable range : 0.1 – 10 ml)

Clinician (midwife) bolus lockout : 1 hour (not adjustable)

Press '**Next**'

Press **'OK**'

Do you want to prime before you start?

Choose 'PRIME' or '**START'**

'**Delivery history**' provides data on PIB and PCEA usage

To change settings

Press '**unlock to patient'** (bottom of screen)

Password **5551**

Press **'View/Edit'** (bottom of screen)

Bolus (0.1-10mls)

Input the required volume

Press '**Ok'**

Press '**Request Continue'** (bottom of screen)

To pause or stop the epidural

Choose '**Press to unlock patient**'

Enter password '**5551**'

Request '**Pause**'

Press '**OK**'

'Press '**Quit**' (bottom left of screen to stop)

<u>Sapphire Epidural 'Quick Tips'</u>

ACTION	***INSTRUCTION***
LOCK	*Lock > Lock Screen > Patient Lockout*
UNLOCK	*SCREEN LOCK: Press to unlock screen > OK* *PATIENT LOCK: Press to unlock patient > enter password > OK*
PAUSE INFUSION	*Request Pause > OK*
RESUME INFUSION	*Request Continue > OK*
REVIEW PROGRAM	*View/Edit > review parameters > OK*
EDIT PROGRAMME	*Request Pause > OK > View/Edit > select parameter and input* *New values > Ok > Ok > Request Continue > Ok*
REVIEW BOLUS HISTORY	*LOCKED SCREEN: Delivery History > Back* *UNLOCKED SCREEN: View/Edit > Delivery History > Exit*
AIR IN LINE	*Mute > Prime > disconnect line from patient > Prime* *Once air has been dispelled from the line > Finish Prime > connect line to the patient > OK*
CHANGE BAG/REPEAT LAST INFUSION	*Mute > Infusion Complete message > OK > Repeat last infusion . review and if correct > OK > Start*
DISCONTINUE INFUSION	*Request Pause > OK > Quit > Quit infusion*
EMERGENCY OFF	*Press and hold on/off button for 5 seconds*
TURN PUMP OFF	*Press STOP press on/off button > OFF*

4. References

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2. Epidural Information Card. Obstetric Anaesthetists Association 2008

3. Van der Vyver M, Halpern S and Joseph G. Patient Controlled epidural analgesia vs continuous infusion for labour analgesia : a meta-analysis. British Journal of Anaesthesia **89** (3): 459-65 (2002)

4. Halpern S, Carvalho B. Patient Controlled Epidural Analgesia for Labour. Anaesthesia & Analgesia **108**: 921-8 (2009)

5. Wong CA, Ratliff JT, Sullivan JT, Scavone BM, Toledo P, Mc Carthy RJ. A Randomised Comparison of Programmed Intermittent Epidural Bolus with Continuous Epidural Infusion for Labour Analgesia. Anaesthesia & Analgesia **102**: 904-9 (2006)

6. Appendix 2 Epidural Analgesia – Patient Controlled Local Operating Procedure (2012). Royal Hospital for Women, NSW, Australia